

[DHS/2022/Q2]

Task 2

Task 2.1

The NDP parade commander needs a program to simulate the parade segment because of a recent proposal for contingent participants to simultaneously enter and leave a confined space on the parade platform.

Using a fixed array, complete the class `CircularQueue` in `task2.ipynb` which is a circular queue with the following methods:

- `enqueue(string)` which takes in a string and adds it to the queue or returns "Cannot enqueue: Queue is full." if the queue is full
- `dequeue` which removes the next item for the queue and returns the value removed

Each item enqueued represents **one participant** from a contingent.

You should **only** edit the code in `task2.ipynb` that is within the two above methods.

For testing purposes, you will be using a queue of size 5. Test your code by using the driver program provided in `task2.ipynb` to produce this output:

```
(1) Enqueue Participant_A, Participant_B, Participant_C, Participant_D.
Fixed array contents:
['Participant_A', 'Participant_B', 'Participant_C', 'Participant_D', None]

(2.1) Deleted value = Participant_A
(2.2) Deleted value = Participant_B
Fixed array contents:
['Participant_A', 'Participant_B', 'Participant_C', 'Participant_D', None]

(3) Enqueue Participant_E, Participant_F, Participant_G.
Fixed array contents:
['Participant_F', 'Participant_G', 'Participant_C', 'Participant_D',
'Participant_E']

(4) Enqueue Participant_H.
Cannot enqueue: Queue is full.
Fixed array contents:
['Participant_F', 'Participant_G', 'Participant_C', 'Participant_D',
'Participant_E']
```

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Task 2.2 (Bonus question)

The parade commander wants to predict the time and name of the contingent entering the platform when the number of participants on the platform crosses a certain overload threshold of 530 people.

Write a function `simulate(filename, threshold)` which takes in a file as `filename` and a positive integer `threshold` which is the maximum number of people the platform should have. `simulate(filename, threshold)` should use `CircularQueue` from Task 2.1 and return a string with information needed by the commander.

You are provided with `marching.csv`. Its first, second, third and fourth columns correspond to each participant's NRIC, contingent name, entry-to-platform time and exit-from-platform time. Each line of `marching.csv` represents **one** person from the contingent.

Test your function with `simulate("marching.csv", 530)` to display:

Platform full at 6:25 pm when RSNMaritimeTrainingandDoctrineCommand is entering.

[5]